MOHAMED KHALIFA

Mechanical Engineering Graduate | Controls, Robotics & Mechatronics

Ottawa, ON | Open to Relocate | (613) 276-5431 | MohamedKhalifa15501@gmail.com

PROFESSIONAL SUMMARY

Mechanical engineering master's graduate with published research in robotic navigation, state estimation and industrial robot control, plus hands-on experience in mechanical design, prototyping and automation. Developed and validated control algorithms in MATLAB/Simulink and integrated sensor-driven systems using SolidWorks, Arduino and PLCs.

TECHNICAL SKILLS

Control & robotics: Unscented Kalman Filter (UKF), sensor fusion, visual-inertial odometry (VIO), PID, computed-torque control, MPC, LQR, state observers, robot kinematics

Mechanical & automation: SolidWorks, 3D CAD, stress/deflection analysis, 3D printing, prototyping, sensor/actuator integration, Arduino, PLC ladder logic, testing and troubleshooting

Software & tools: MATLAB/Simulink, Simscape, Python, C/C++, LabVIEW, Git, ROS 2, Nvidia Isaac Sim (fundamentals)

RESEARCH & PROFESSIONAL EXPERIENCE

Master's Thesis Research — Robotics & Navigation | Carleton University | *Sep 2024 – May 2026*

- Developed a dual quaternion-based Unscented Kalman Filter to estimate drone position and orientation from stereo camera and inertial data for GPS-denied navigation in MATLAB.
- Evaluated on three recorded drone-flight sequences; achieved 0.26 m position RMSE versus 0.33 m for a quaternion UKF baseline on the most challenging flight.

Engineering Teaching Assistant | Carleton University | *Sep 2024 – Apr 2026*

- Led problem-solving and lab sessions in Feedback Control, Engineering Dynamics, Mechanical Design and Fluid Mechanics.

Robotics Research Intern | Intelligent Manufacturing & Robotics Research Center, KFUPM | *Jun – Aug 2023*

- Modeled a 3-DOF robotic drilling arm in SolidWorks and MATLAB/Simulink; developed its kinematics, dynamics, and trajectory planning.
- Implemented PID and computed-torque controllers for a 3-DOF robotic drilling system in MATLAB/Simulink; reduced end-effector position tracking RMSE by 98% (X) and 94% (Y) versus PID.

SELECTED ENGINEERING PROJECTS

Model Predictive Control & State Observer | Graduate Project, Carleton University | *2024 – 2025*

- Developed constrained MPC in MATLAB with a 20-step prediction horizon, 5-step control horizon and 50 ms sampling; expanded actuator limits from ±10 N to ±15 N to meet cart-position (±1 m) and tilt (±45°) constraints in simulation. Implemented LQR with a state observer for comparison against the MPC approach.

Automated Five-Stage Water Filtration System | Senior Capstone, KFUPM | *2023*

- Built a five-stage automated filtration prototype integrating four sensor types (pressure, temperature, flow, and level) with Arduino Mega-controlled pumps and valves.
- Led procurement, mechanical assembly, and wiring; resolved fluid leaks using threaded fittings and silicone sealant, and troubleshot relay operation.

Mechanical Design & Robotics | KFUPM | *2022 – 2023*

- Designed and 3D-printed a 2-DOF pick-and-place robot; programmed PLC conveyor sorting; analyzed a robotic gripper against a 150 kg design-load requirement and a <5 kg target mass.

EDUCATION

Master of Applied Science (MASc), Mechanical Engineering — *09/2024 – 05/2026*
Carleton University, Ottawa, Canada — GPA: 11.6 / 12

Bachelor of Science, Mechanical Engineering — *09/2019 – 01/2024*
King Fahd University of Petroleum and Minerals, Dhahran, Saudi Arabia — GPA: 3.921 / 4

PUBLICATIONS & PATENT APPLICATIONS

Measurement (2026) — Dual Quaternion-Based Unscented Kalman Filter with Visual Inertial Odometry for Navigation in GPS-Denied Environments (first author).

IEEE ICETSIS (2024) — Position and Torque Joint Control for Precision Drilling in Robotic Manufacturing Applications (first author).

Arabian Journal for Science and Engineering (2025) — Actuated Rotational Adaptive Optics (co-authored review).

Patent applications (3) — Co-inventor on two U.S. membrane-distillation inventions and one Saudi vehicular desalination invention.